File Transfers

There are many situations when transferring files to or from a target system is necessary. Let's imagine the following scenario:

Setting the Stage

During an engagement, we gain remote code execution (RCE) on an IIS web server via an unrestricted file upload vulnerability. We upload a web shell initially and then send ourselves a reverse shell to enumerate the system further in an attempt to escalate privileges. We attempt to use PowerShell to transfer **PowerUp.ps1** (a PowerShell script to enumerate privilege escalation vectors), but PowerShell is blocked by the **Application Control Policy**. We perform our local enumeration manually and find that we have **SeImpersonatePrivilege**. We need to transfer a binary to our target machine to escalate privileges using the **PrintSpoofer** tool. We then try to use **Certutil** to download the file we compiled ourselves directly from our own GitHub, but the organization has strong web content filtering in place. We cannot access websites such as GitHub, Dropbox, Google Drive, etc., that can be used to transfer files. Next, we set up an FTP Server and tried to use the Windows FTP client to transfer files, but the network firewall blocked outbound traffic for port 21 (TCP). We tried to use the **Impacket smbserver** tool to create a folder, and we found that outgoing traffic to TCP port 445 (SMB) was allowed. We used this file transfer method to successfully copy the binary onto our target machine and accomplish our goal of escalating privileges to an administrator-level user.

Understanding different ways to perform file transfers and how networks operate can help us accomplish our goals during an assessment. We must be aware of host controls that may prevent our actions, like application whitelisting or AV/EDR blocking specific applications or activities. File transfers are also affected by network devices such as Firewalls, IDS, or IPS which can monitor or block particular ports or uncommon operations.

File transfer is a core feature of any operating system, and many tools exist to achieve this. However, many of these tools may be blocked or monitored by diligent administrators, and it is worth reviewing a range of techniques that may be possible in a given environment.

This module covers techniques that leverage tools and applications commonly available on Windows and Linux systems. The list of techniques is not exhaustive. The information within this module can also be used as a reference guide when working through other HTB Academy modules, as many of the in-module exercises will require us to transfer files to/from a target host or to/from the provided Pwnbox. Target Windows and Linux machines are provided to complete a few hands-on exercises as part of the module. It is worth utilizing these targets to experiment with as many of the techniques demonstrated in the module sections as possible. Observe the nuances between the different transfer methods and note down situations where they would be helpful. Once you have completed this module, try out the various techniques in other HTB Academy modules and boxes and labs on the HTB main platform.